

Final project

July 26, 2024

```
[3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

```
[4]: df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')
```

```
[5]: df.shape
```

```
[5]: (1470, 35)
```

```
[6]: df.head()
```

```
[6]:   Age Attrition   BusinessTravel   DailyRate   Department \
0   41      Yes   Travel_Rarely      1102      Sales
1   49      No   Travel_Frequently      279  Research & Development
2   37      Yes   Travel_Rarely      1373  Research & Development
3   33      No   Travel_Frequently      1392  Research & Development
4   27      No   Travel_Rarely      591   Research & Development

   DistanceFromHome   Education   EducationField   EmployeeCount   EmployeeNumber \
0                1           2   Life Sciences           1           1
1                8           1   Life Sciences           1           2
2                2           2         Other           1           4
3                3           4   Life Sciences           1           5
4                2           1         Medical           1           7

   ...   RelationshipSatisfaction   StandardHours   StockOptionLevel \
0   ...                1           80           0
1   ...                4           80           1
2   ...                2           80           0
3   ...                3           80           0
4   ...                4           80           1

   TotalWorkingYears   TrainingTimesLastYear   WorkLifeBalance   YearsAtCompany \
0                8           0           1           6
1               10           3           3          10
```

2	7	3	3	0
3	8	3	3	8
4	6	3	3	2

	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
0	4	0	5
1	7	1	7
2	0	0	0
3	7	3	0
4	2	2	2

[5 rows x 35 columns]

```
[7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1470 non-null   int64
1   Attrition                           1470 non-null   object
2   BusinessTravel                       1470 non-null   object
3   DailyRate                           1470 non-null   int64
4   Department                           1470 non-null   object
5   DistanceFromHome                    1470 non-null   int64
6   Education                           1470 non-null   int64
7   EducationField                       1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                              1470 non-null   object
12  HourlyRate                           1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                              1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                       1470 non-null   object
18  MonthlyIncome                       1470 non-null   int64
19  MonthlyRate                         1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
21  Over18                              1470 non-null   object
22  OverTime                            1470 non-null   object
23  PercentSalaryHike                   1470 non-null   int64
24  PerformanceRating                   1470 non-null   int64
25  RelationshipSatisfaction             1470 non-null   int64
26  StandardHours                       1470 non-null   int64
27  StockOptionLevel                    1470 non-null   int64
```

28	TotalWorkingYears	1470	non-null	int64
29	TrainingTimesLastYear	1470	non-null	int64
30	WorkLifeBalance	1470	non-null	int64
31	YearsAtCompany	1470	non-null	int64
32	YearsInCurrentRole	1470	non-null	int64
33	YearsSinceLastPromotion	1470	non-null	int64
34	YearsWithCurrManager	1470	non-null	int64

dtypes: int64(26), object(9)
memory usage: 402.1+ KB

```
[8]: pd.isnull(df).sum()
```

```
[8]: Age                                0
Attrition                             0
BusinessTravel                         0
DailyRate                             0
Department                             0
DistanceFromHome                       0
Education                             0
EducationField                         0
EmployeeCount                          0
EmployeeNumber                         0
EnvironmentSatisfaction                0
Gender                                 0
HourlyRate                             0
JobInvolvement                         0
JobLevel                              0
JobRole                               0
JobSatisfaction                        0
MaritalStatus                         0
MonthlyIncome                         0
MonthlyRate                           0
NumCompaniesWorked                    0
Over18                                0
OverTime                              0
PercentSalaryHike                     0
PerformanceRating                     0
RelationshipSatisfaction               0
StandardHours                         0
StockOptionLevel                      0
TotalWorkingYears                     0
TrainingTimesLastYear                 0
WorkLifeBalance                       0
YearsAtCompany                        0
YearsInCurrentRole                    0
YearsSinceLastPromotion               0
YearsWithCurrManager                  0
```

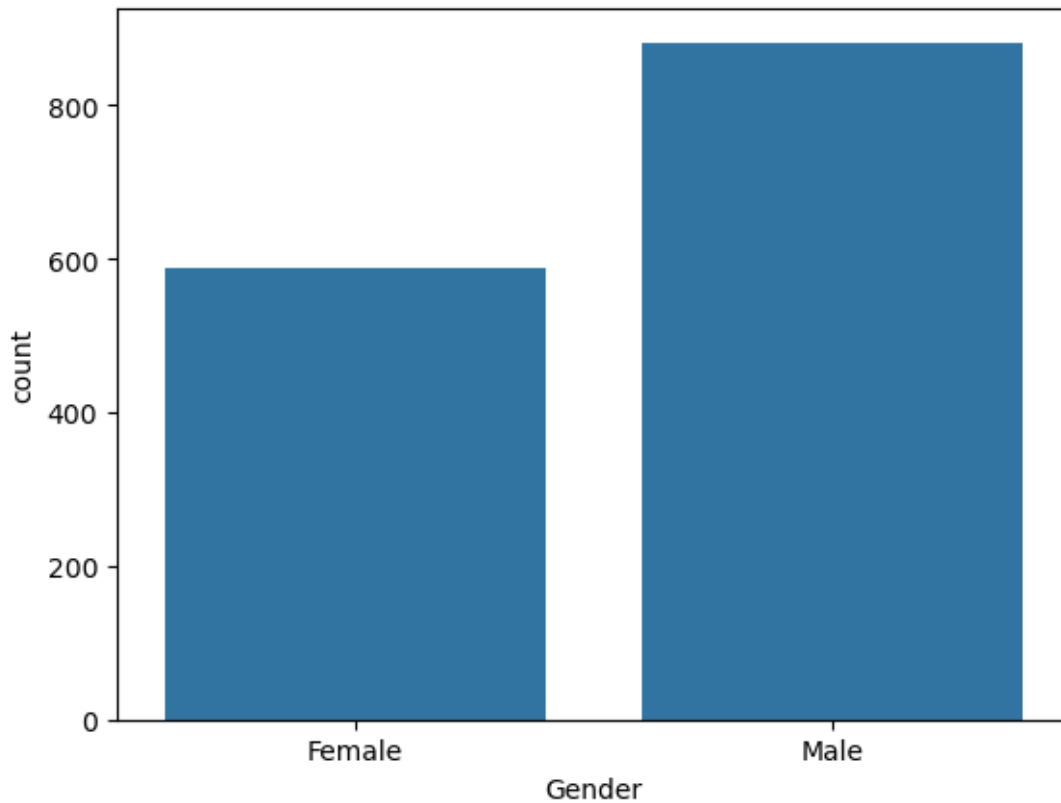
dtype: int64

```
[9]: df.columns
```

```
[9]: Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',  
        'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',  
        'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',  
        'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',  
        'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',  
        'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',  
        'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',  
        'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',  
        'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',  
        'YearsWithCurrManager'],  
        dtype='object')
```

```
[24]: ax = sns.countplot(x='Gender', data=df)  
  
for bars in ax.container:  
    ax.bar_label(bars)
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[24], line 3  
      1 ax = sns.countplot(x='Gender', data=df)  
----> 3 for bars in ax.container:  
      4     ax.bar_label(bars)  
  
AttributeError: 'Axes' object has no attribute 'container'
```



```
[27]: df[['DailyRate', 'Department']]
```

```
[27]:
```

	DailyRate	Department
0	1102	Sales
1	279	Research & Development
2	1373	Research & Development
3	1392	Research & Development
4	591	Research & Development
...	...	...
1465	884	Research & Development
1466	613	Research & Development
1467	155	Research & Development
1468	1023	Sales
1469	628	Research & Development

```
[1470 rows x 2 columns]
```

```
[17]: df[['EducationField', 'EmployeeCount', 'EmployeeNumber']]
```

```
[17]:
```

	EducationField	EmployeeCount	EmployeeNumber
0	Life Sciences	1	1

1	Life Sciences	1	2
2	Other	1	4
3	Life Sciences	1	5
4	Medical	1	7
...	...	...	...
1465	Medical	1	2061
1466	Medical	1	2062
1467	Life Sciences	1	2064
1468	Medical	1	2065
1469	Medical	1	2068

[1470 rows x 3 columns]

```
[18]: df[['EnvironmentSatisfaction', 'Gender', 'HourlyRate', 'JobInvolvement']]
```

```
[18]:
```

	EnvironmentSatisfaction	Gender	HourlyRate	JobInvolvement
0	2	Female	94	3
1	3	Male	61	2
2	4	Male	92	2
3	4	Female	56	3
4	1	Male	40	3
...	...	...	...	...
1465	3	Male	41	4
1466	4	Male	42	2
1467	2	Male	87	4
1468	4	Male	63	2
1469	2	Male	82	4

[1470 rows x 4 columns]

```
[19]: df[['JobLevel', 'JobRole', 'JobSatisfaction',
'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
'RelationshipSatisfaction']]
```

```
[19]:
```

	JobLevel	JobRole	JobSatisfaction	MaritalStatus	\
0	2	Sales Executive	4	Single	
1	2	Research Scientist	2	Married	
2	1	Laboratory Technician	3	Single	
3	1	Research Scientist	3	Married	
4	1	Laboratory Technician	2	Married	
...	...	...	...	...	
1465	2	Laboratory Technician	4	Married	
1466	3	Healthcare Representative	1	Married	
1467	2	Manufacturing Director	2	Married	
1468	2	Sales Executive	2	Married	
1469	2	Laboratory Technician	3	Married	

	MonthlyIncome	MonthlyRate	NumCompaniesWorked	Over18	OverTime	\
0	5993	19479	8	Y	Yes	
1	5130	24907	1	Y	No	
2	2090	2396	6	Y	Yes	
3	2909	23159	1	Y	Yes	
4	3468	16632	9	Y	No	
...	...	...	...	...	...	
1465	2571	12290	4	Y	No	
1466	9991	21457	4	Y	No	
1467	6142	5174	1	Y	Yes	
1468	5390	13243	2	Y	No	
1469	4404	10228	2	Y	No	

	PercentSalaryHike	PerformanceRating	RelationshipSatisfaction
0	11	3	1
1	23	4	4
2	15	3	2
3	11	3	3
4	12	3	4
...	...	...	...
1465	17	3	3
1466	15	3	1
1467	20	4	2
1468	14	3	4
1469	12	3	1

[1470 rows x 12 columns]

```
[20]: df[['StandardHours', 'StockOptionLevel',
          'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
          'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
          'YearsWithCurrManager']]
```

	StandardHours	StockOptionLevel	TotalWorkingYears	\
0	80	0	8	
1	80	1	10	
2	80	0	7	
3	80	0	8	
4	80	1	6	
...	...	...	...	
1465	80	1	17	
1466	80	1	9	
1467	80	1	6	
1468	80	0	17	
1469	80	0	6	

	TrainingTimesLastYear	WorkLifeBalance	YearsAtCompany	\
0	0	1	6	
1	3	3	10	
2	3	3	0	
3	3	3	8	
4	3	3	2	
...	...	...	...	
1465	3	3	5	
1466	5	3	7	
1467	0	3	6	
1468	3	2	9	
1469	3	4	4	

	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
0	4	0	5
1	7	1	7
2	0	0	0
3	7	3	0
4	2	2	2
...	...	...	...
1465	2	0	3
1466	7	1	7
1467	2	0	3
1468	6	0	8
1469	3	1	2

[1470 rows x 9 columns]

```
[21]: df.drop(labels='StockOptionLevel',axis=1)
```

```
[21]:
```

	Age	Attrition	BusinessTravel	DailyRate	Department	\
0	41	Yes	Travel_Rarely	1102	Sales	
1	49	No	Travel_Frequently	279	Research & Development	
2	37	Yes	Travel_Rarely	1373	Research & Development	
3	33	No	Travel_Frequently	1392	Research & Development	
4	27	No	Travel_Rarely	591	Research & Development	
...	...	...	...	...	...	
1465	36	No	Travel_Frequently	884	Research & Development	
1466	39	No	Travel_Rarely	613	Research & Development	
1467	27	No	Travel_Rarely	155	Research & Development	
1468	49	No	Travel_Frequently	1023	Sales	
1469	34	No	Travel_Rarely	628	Research & Development	

	DistanceFromHome	Education	EducationField	EmployeeCount	\
0	1	2	Life Sciences	1	
1	8	1	Life Sciences	1	
2	2	2	Other	1	

3	3	4	Life Sciences	1
4	2	1	Medical	1
...	...	...	...	...
1465	23	2	Medical	1
1466	6	1	Medical	1
1467	4	3	Life Sciences	1
1468	2	3	Medical	1
1469	8	3	Medical	1

	EmployeeNumber	...	PerformanceRating	RelationshipSatisfaction	\
0	1	...	3	1	
1	2	...	4	4	
2	4	...	3	2	
3	5	...	3	3	
4	7	...	3	4	
...	...	...	...	...	
1465	2061	...	3	3	
1466	2062	...	3	1	
1467	2064	...	4	2	
1468	2065	...	3	4	
1469	2068	...	3	1	

	StandardHours	TotalWorkingYears	TrainingTimesLastYear	WorkLifeBalance	\
0	80	8	0	1	
1	80	10	3	3	
2	80	7	3	3	
3	80	8	3	3	
4	80	6	3	3	
...	...	...	...	...	
1465	80	17	3	3	
1466	80	9	5	3	
1467	80	6	0	3	
1468	80	17	3	2	
1469	80	6	3	4	

	YearsAtCompany	YearsInCurrentRole	YearsSinceLastPromotion	\
0	6	4	0	
1	10	7	1	
2	0	0	0	
3	8	7	3	
4	2	2	2	
...	...	...	...	
1465	5	2	0	
1466	7	7	1	
1467	6	2	0	
1468	9	6	0	
1469	4	3	1	

	YearsWithCurrManager
0	5
1	7
2	0
3	0
4	2
...	...
1465	3
1466	7
1467	3
1468	8
1469	2

[1470 rows x 34 columns]

```
[23]: df.describe()
```

```
[23]:
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	\
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	
mean	36.923810	802.485714	9.192517	2.912925	1.0	
std	9.135373	403.509100	8.106864	1.024165	0.0	
min	18.000000	102.000000	1.000000	1.000000	1.0	
25%	30.000000	465.000000	2.000000	2.000000	1.0	
50%	36.000000	802.000000	7.000000	3.000000	1.0	
75%	43.000000	1157.000000	14.000000	4.000000	1.0	
max	60.000000	1499.000000	29.000000	5.000000	1.0	

	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	\
count	1470.000000	1470.000000	1470.000000	1470.000000	
mean	1024.865306	2.721769	65.891156	2.729932	
std	602.024335	1.093082	20.329428	0.711561	
min	1.000000	1.000000	30.000000	1.000000	
25%	491.250000	2.000000	48.000000	2.000000	
50%	1020.500000	3.000000	66.000000	3.000000	
75%	1555.750000	4.000000	83.750000	3.000000	
max	2068.000000	4.000000	100.000000	4.000000	

	JobLevel	...	RelationshipSatisfaction	StandardHours	\
count	1470.000000	...	1470.000000	1470.0	
mean	2.063946	...	2.712245	80.0	
std	1.106940	...	1.081209	0.0	
min	1.000000	...	1.000000	80.0	
25%	1.000000	...	2.000000	80.0	
50%	2.000000	...	3.000000	80.0	
75%	3.000000	...	4.000000	80.0	
max	5.000000	...	4.000000	80.0	

	StockOptionLevel	TotalWorkingYears	TrainingTimesLastYear \
count	1470.000000	1470.000000	1470.000000
mean	0.793878	11.279592	2.799320
std	0.852077	7.780782	1.289271
min	0.000000	0.000000	0.000000
25%	0.000000	6.000000	2.000000
50%	1.000000	10.000000	3.000000
75%	1.000000	15.000000	3.000000
max	3.000000	40.000000	6.000000

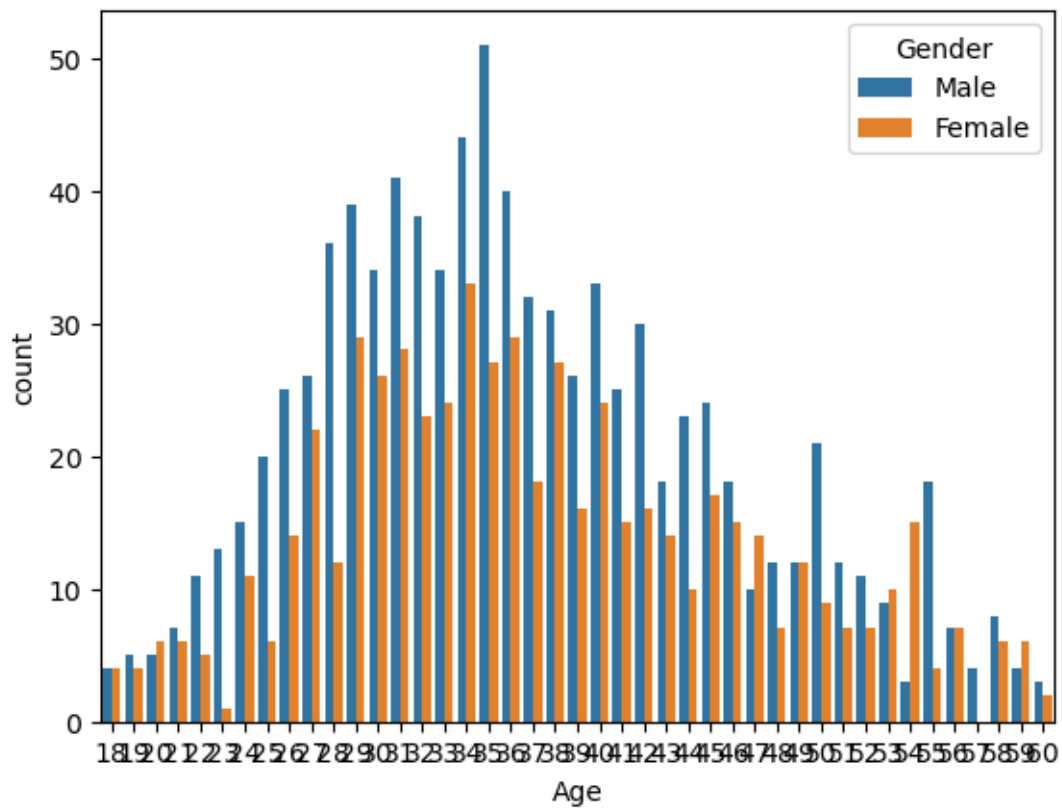
	WorkLifeBalance	YearsAtCompany	YearsInCurrentRole \
count	1470.000000	1470.000000	1470.000000
mean	2.761224	7.008163	4.229252
std	0.706476	6.126525	3.623137
min	1.000000	0.000000	0.000000
25%	2.000000	3.000000	2.000000
50%	3.000000	5.000000	3.000000
75%	3.000000	9.000000	7.000000
max	4.000000	40.000000	18.000000

	YearsSinceLastPromotion	YearsWithCurrManager
count	1470.000000	1470.000000
mean	2.187755	4.123129
std	3.222430	3.568136
min	0.000000	0.000000
25%	0.000000	2.000000
50%	1.000000	3.000000
75%	3.000000	7.000000
max	15.000000	17.000000

[8 rows x 26 columns]

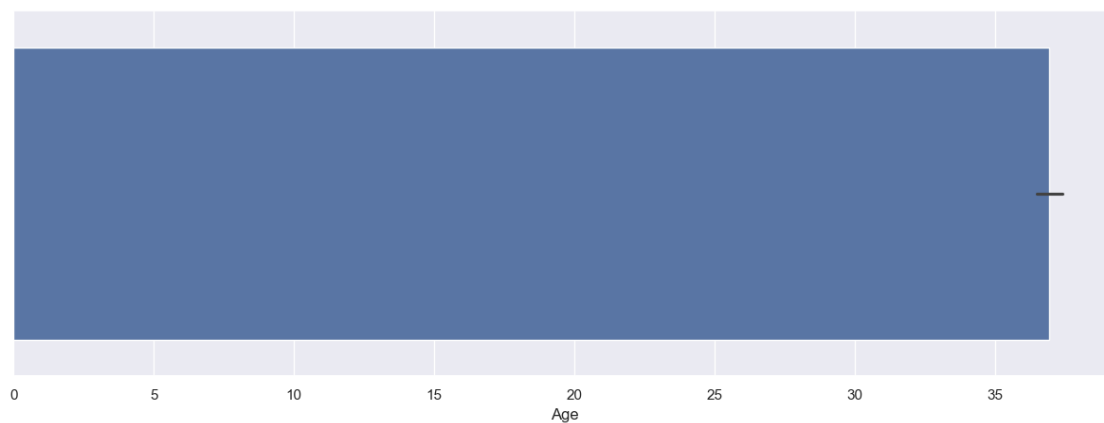
```
[30]: sns.countplot(data = df,x = 'Age',hue = 'Gender')
```

```
[30]: <Axes: xlabel='Age', ylabel='count'>
```



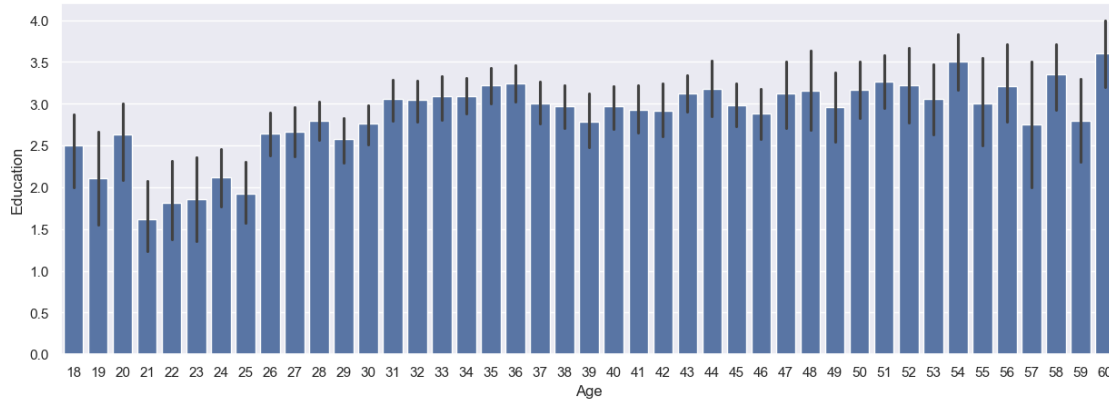
```
[46]: sns.set(rc={'figure.figsize':(15,5)})
      sns.barplot(x='Age',data = df )
```

```
[46]: <Axes: xlabel='Age'>
```



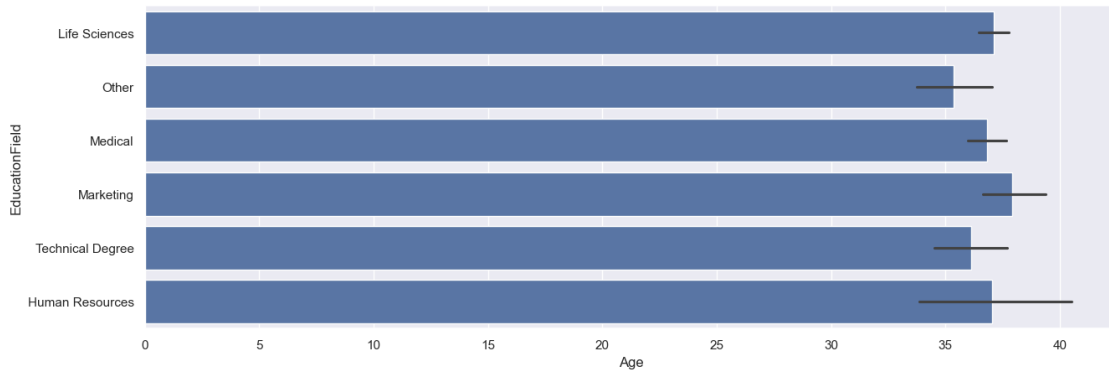
```
[45]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x='Age',y='Education',data = df )
```

```
[45]: <Axes: xlabel='Age', ylabel='Education'>
```



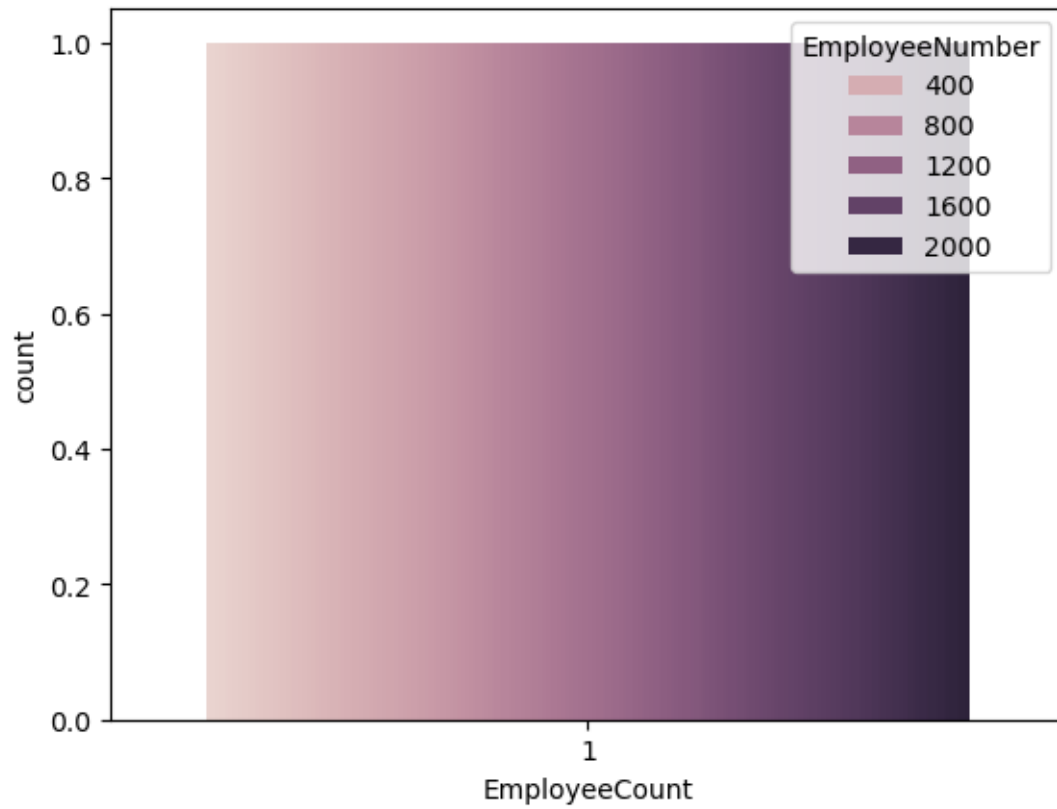
```
[44]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x='Age',y='EducationField',data = df )
```

```
[44]: <Axes: xlabel='Age', ylabel='EducationField'>
```



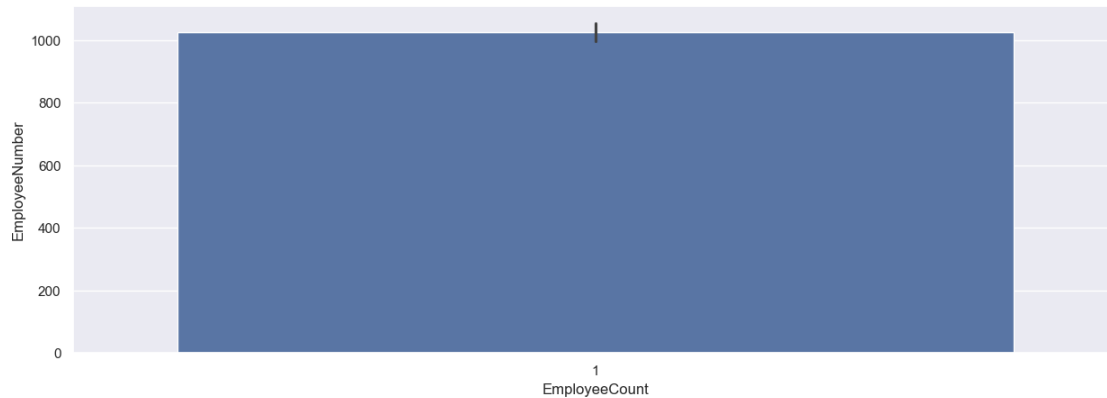
```
[35]: sns.countplot(data = df,x = 'EmployeeCount',hue = 'EmployeeNumber')
```

```
[35]: <Axes: xlabel='EmployeeCount', ylabel='count'>
```



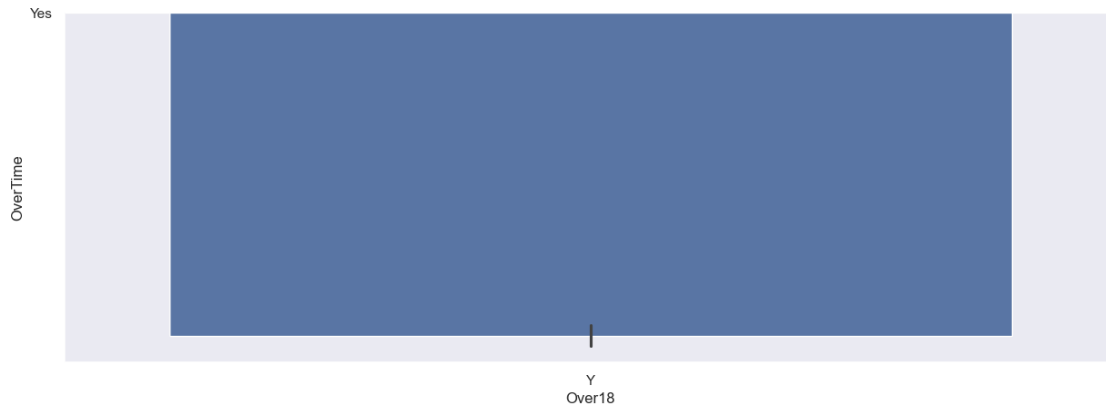
```
[42]: sns.set(rc={'figure.figsize':(15,5)})
      sns.barplot(x='EmployeeCount',y='EmployeeNumber',data = df )
```

```
[42]: <Axes: xlabel='EmployeeCount', ylabel='EmployeeNumber'>
```



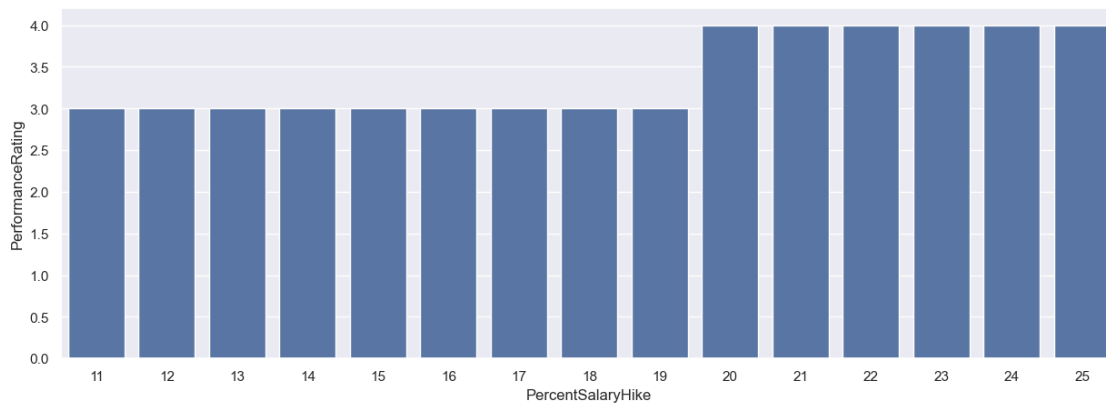
```
[47]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x = 'Over18', y = 'OverTime', data = df )
```

```
[47]: <Axes: xlabel='Over18', ylabel='OverTime'>
```



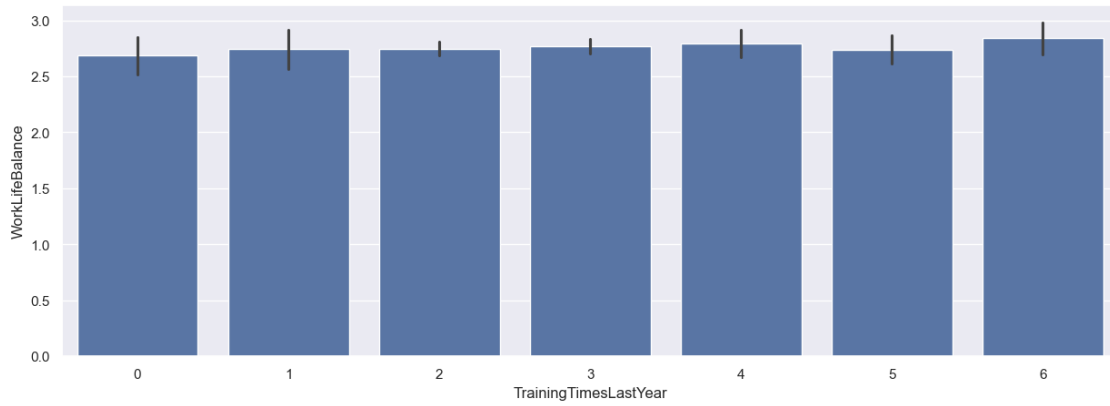
```
[48]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x = 'PercentSalaryHike', y = 'PerformanceRating', data = df )
```

```
[48]: <Axes: xlabel='PercentSalaryHike', ylabel='PerformanceRating'>
```



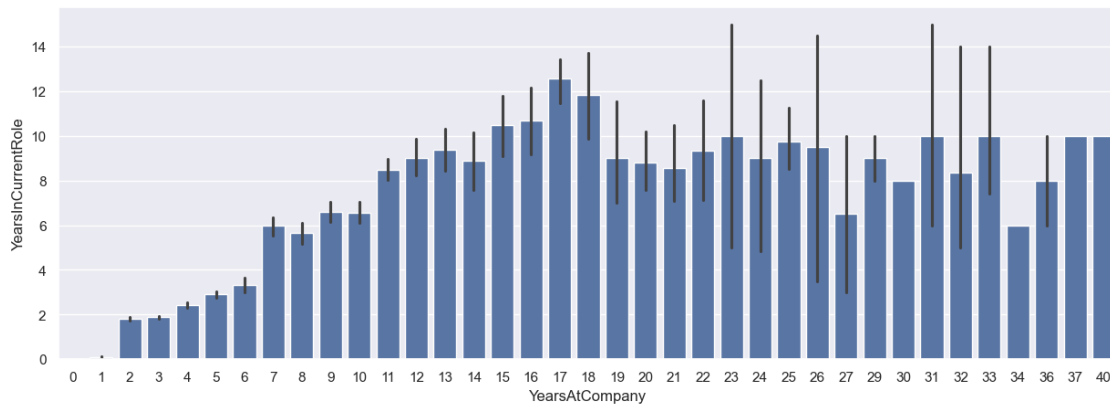
```
[49]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x = 'TrainingTimesLastYear', y = 'WorkLifeBalance', data = df )
```

```
[49]: <Axes: xlabel='TrainingTimesLastYear', ylabel='WorkLifeBalance'>
```



```
[54]: sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(x = 'YearsAtCompany', y= 'YearsInCurrentRole',data = df )
```

```
[54]: <Axes: xlabel='YearsAtCompany', ylabel='YearsInCurrentRole'>
```



```
[ ]:
```